Measuring Agenda-Setting Influence from Legislative Speech

Jack Blumenau

University College London

June 22nd, 2018

Motivation

Ken Clarke, MP for Rushcliffe

"A masterclass display of oratory and expertise."

Methodological argument

Agenda-setting influence: the ability of a speaker to mould discussion of an issue towards their own framing

- How do speakers attribute importance to one another?
  1. Explicitly
  2. Implicitly
- Adopting the framing of an issue implies a perspective is important
- Influential MP's 'shape the debate' by changing the vocabulary of other MPs
- Implication: infer influence from patterns of word use in debate

Penalised regression for modelling influence

- Represent each speech as a vector of word proportions, w_i
- Assumption: If w_i is predictive of w_j, then i influences j
- Problem: relationship between i and j might be confounded by h
- Solution: model each speech as a function of all previous speeches

$$w_j = \sum_{s=1}^{j-1} \beta_s w_s + \epsilon_j$$

- β_i captures the influence of i on j conditional on all other speeches
- Penalise β_s via LASSO $\rightarrow$ 'hard' for one speech to influence another
- Characterise debate as a matrix of summed β_s between speaker-pairs

Ranking influential speakers

Given these patterns of influence, how should we rank the MPs?

$$
\begin{array}{c c}
 & \begin{array}{ccccc} B1 & B2 & B3 & PM & B4 \end{array} \\
\begin{array}{c} B1 \\ B2 \\ B3 \\ PM \\ B4 \end{array} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}
\end{array}
$$

Ranking influential speakers

Degree rank: MPs who influence other MPs are influential

$$
\begin{array}{c c c}
 & \begin{array}{ccccc} B1 & B2 & B3 & PM & B4 \end{array} & \text{Degree Rank} \\
\begin{array}{c} B1 \\ B2 \\ B3 \\ PM \\ B4 \end{array} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} & \begin{array}{c} 2 \\ 3 \\ 3 \\ 1 \\ 3 \end{array}
\end{array}
$$

Ranking influential speakers

Degree rank: MPs who influence other MPs are influential

$$
\begin{array}{c c c}
 & \begin{array}{ccccc} B1 & B2 & B3 & PM & B4 \end{array} & \text{Degree Rank} \\
\begin{array}{c} B1 \\ B2 \\ B3 \\ PM \\ B4 \end{array} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} & \begin{array}{c} 2 \\ 3 \\ 3 \\ 1 \\ 3 \end{array}
\end{array}
$$

Ranking influential speakers

Degree rank: MPs who influence other MPs are influential

$$
\begin{array}{c c c}
 & \begin{array}{ccccc} B1 & B2 & B3 & PM & B4 \end{array} & \text{Degree Rank} \\
\begin{array}{c} B1 \\ B2 \\ B3 \\ PM \\ B4 \end{array} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} & \begin{array}{c} 2 \\ 3 \\ 3 \\ 1 \\ 3 \end{array}
\end{array}
$$

Ranking influential speakers

PageRank: MPs who influence other *influential* MPs are influential

$$
\begin{array}{c c}
 & \begin{array}{ccccc} B1 & B2 & B3 & PM & B4 \end{array} \\
\begin{array}{c} B1 \\ B2 \\ B3 \\ PM \\ B4 \end{array} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}
\end{array}
$$

Degree Rank	*PageRank*
2	3
3	2
3	5
1	1
3	4

Ranking influential speakers

PageRank: MPs who influence other *influential* MPs are influential

$$
\begin{array}{c c}
 & \begin{array}{ccccc} B1 & B2 & B3 & PM & B4 \end{array} \\
\begin{array}{c} B1 \\ B2 \\ B3 \\ PM \\ B4 \end{array} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}
\end{array}
$$

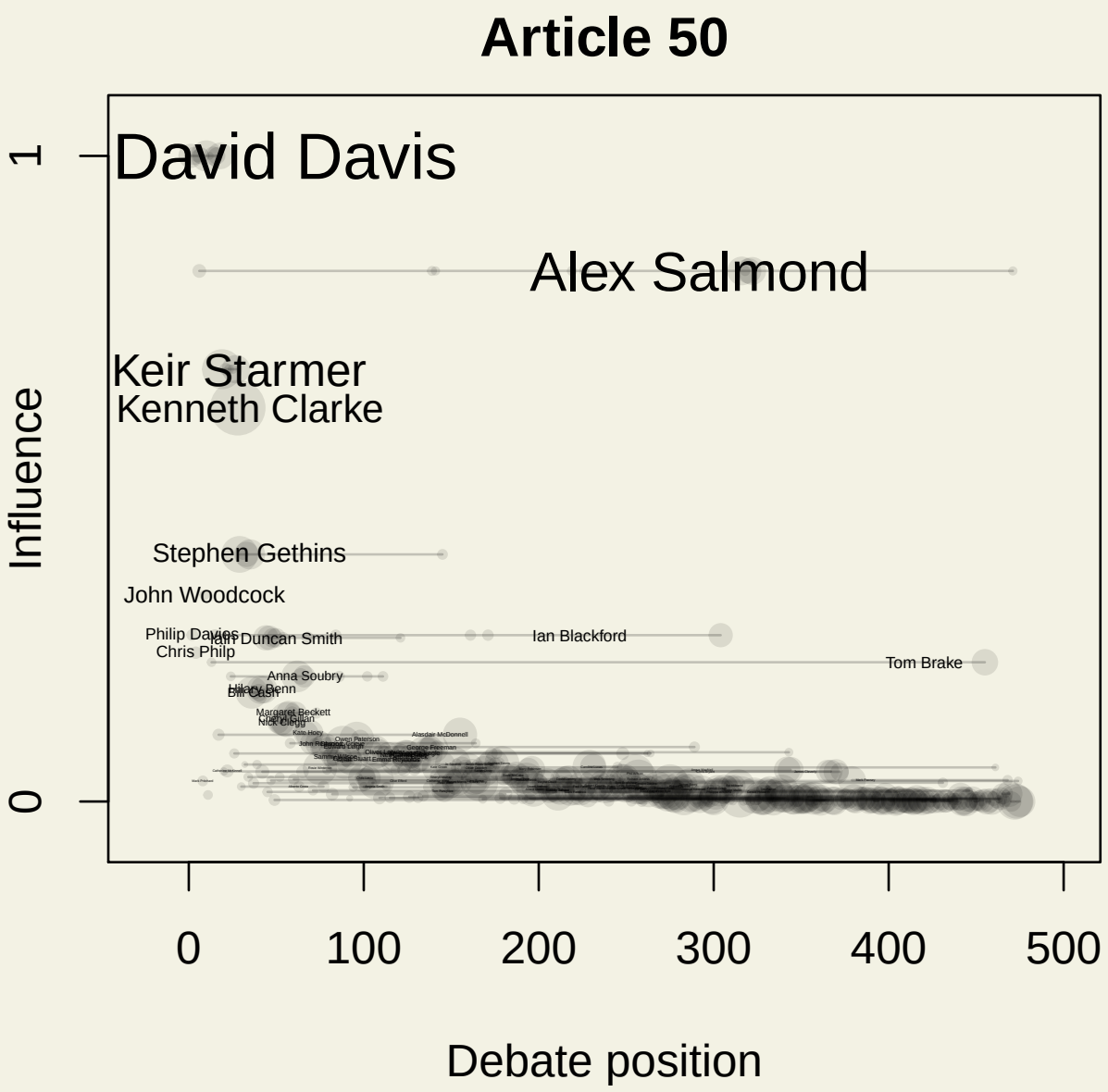
	Degree Rank	*PageRank*
$B1$	2	3
$B2$	3	2
$B3$	3	5
PM	1	1
$B4$	3	4

$\rightarrow$ rank MPs by the *quantity* & *quality* of those they influence

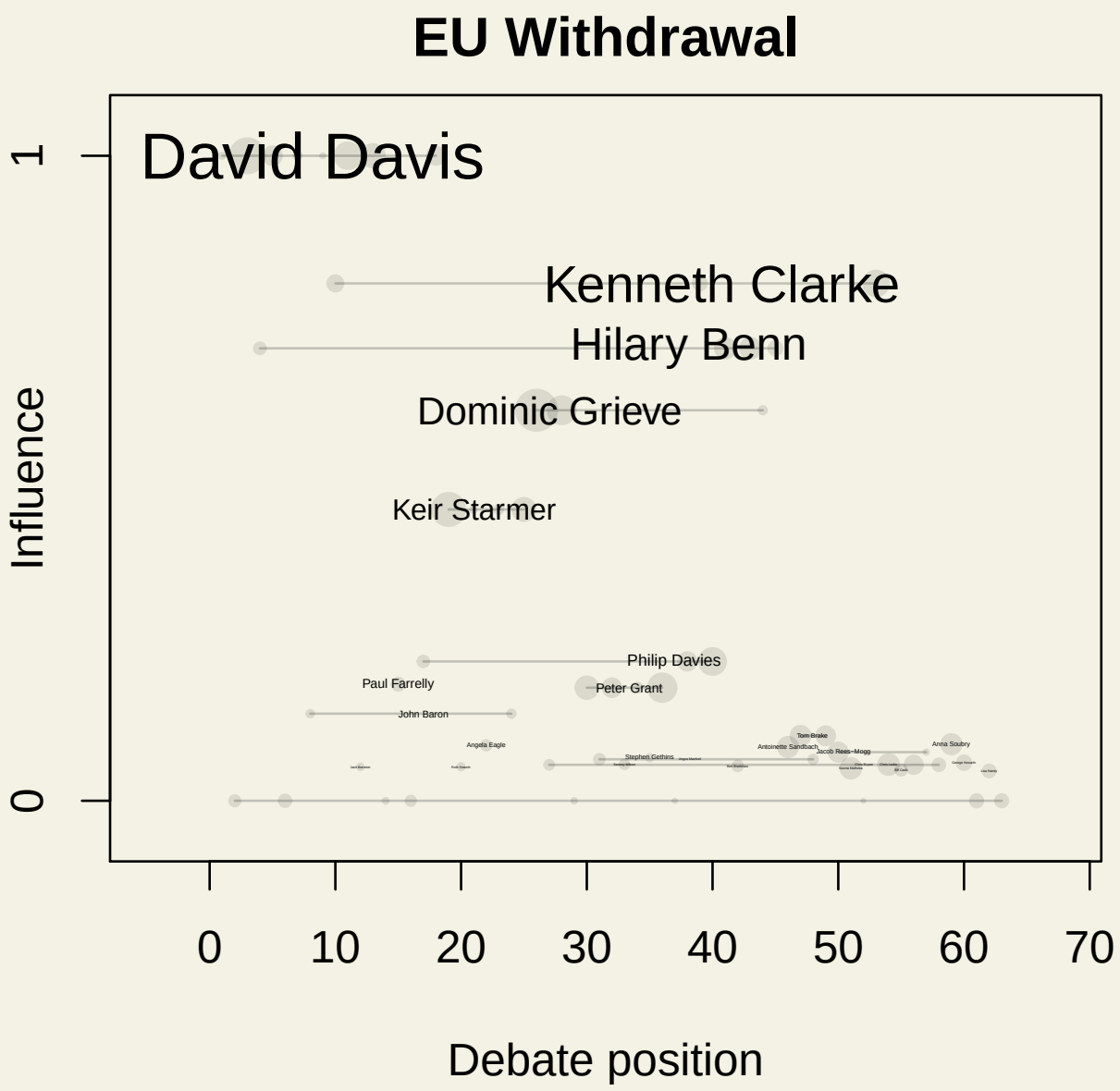
Data

- All debates in the UK House of Commons between 1979 and 2018
- $\approx$ 60,000 debates
- $\approx$ 1.5 million speeches
- For each debate, score the relative influence of each MP via LASSO

Example debate



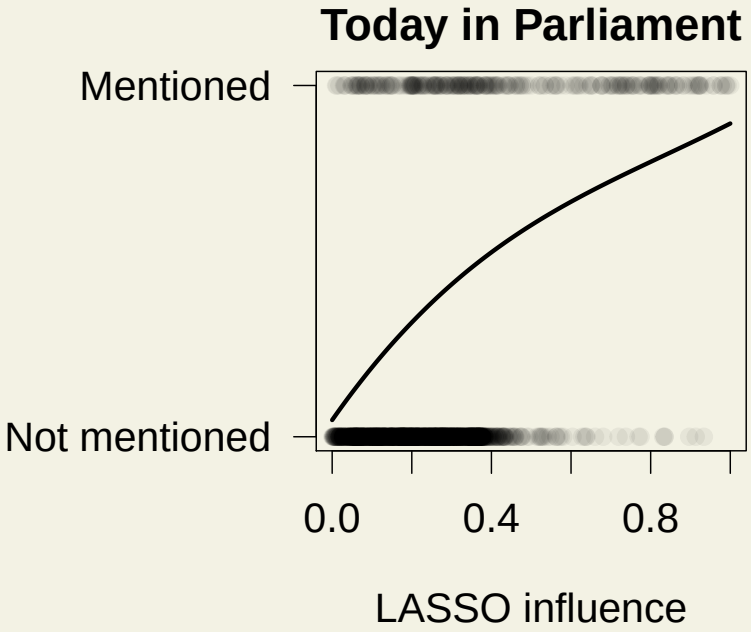
Example debate



Debate-level validation

Do influential **speeches** receive more media coverage?

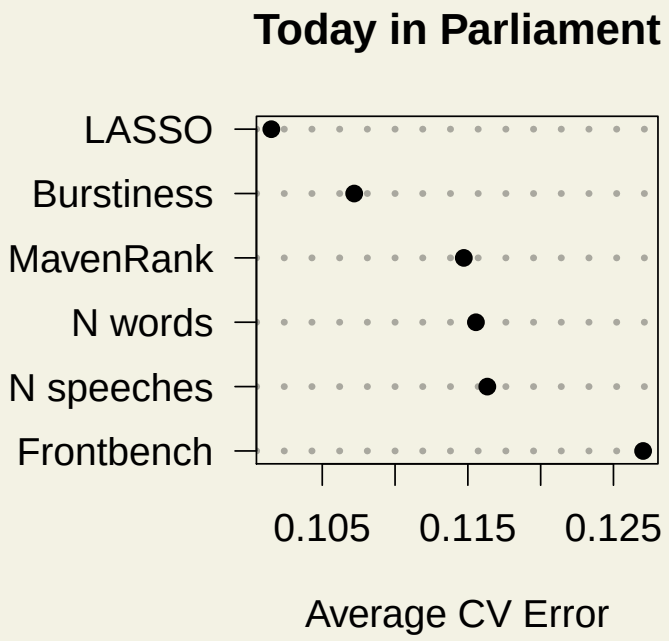
- BBC's 'Today in Parliament' radio programme
- Excerpts from the most important speeches each day
- 17 days in 2018, 78 debates, 1749 speaker-debate observations



Debate-level validation

Do influential **speeches** receive more media coverage?

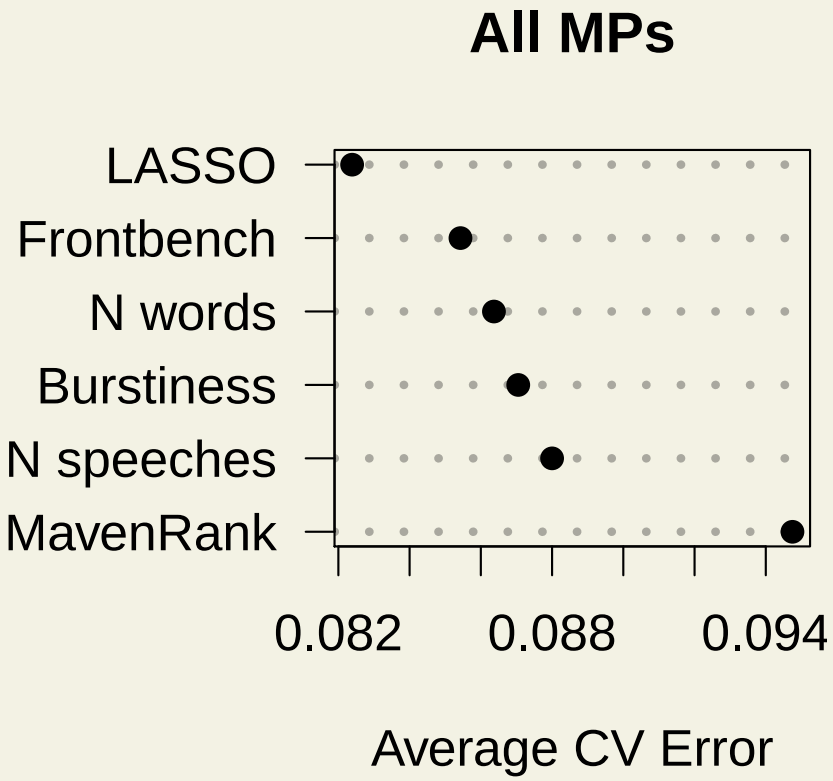
- BBC's 'Today in Parliament' radio programme
- Excerpts from the most important speeches each day
- 17 days in 2018, 78 debates, 1749 speaker-debate observations



Individual-level validation

Do influential **speakers** receive more media coverage?

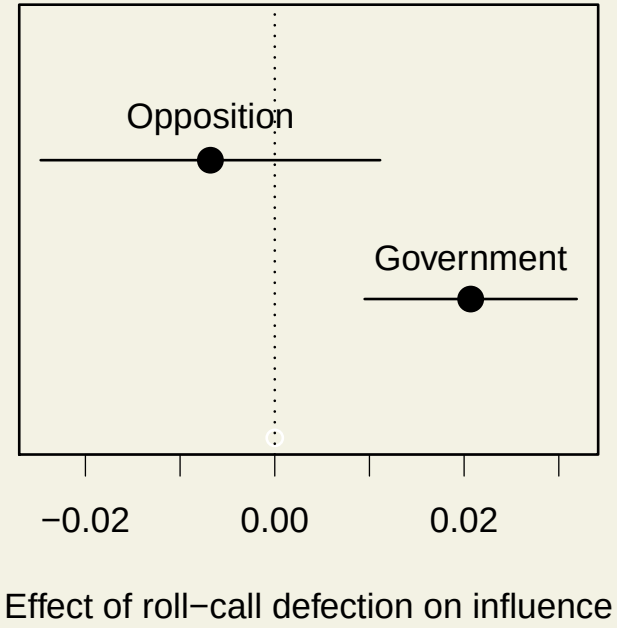
- For each year, measure 'power' of MPs from newspaper mentions (Ban et al, 2017)
- 19,000 Guardian articles from 2010 to 2017
- For each year, measure relative influence in debate of MPs
- Correlation of 0.42 between influence and Guardian mentions



Influence and roll-call defections

Are government rebels influential?

- Government rebels are more likely to give speeches to signal opposition (Slapin et al, 2018)
- Match debates to parliamentary votes: 42,000 speaker-debate observations
- Government rebels are $\approx 10\%$ more influential than loyalists



Discussion

- New measure of agenda-setting influence based on diffusion of word use throughout a debate
- Extensive validation at debate- and individual-levels
- Applications to important questions in legislative politics
- Future work will focus on quantifying uncertainty in the influence scores, and in better/other applications